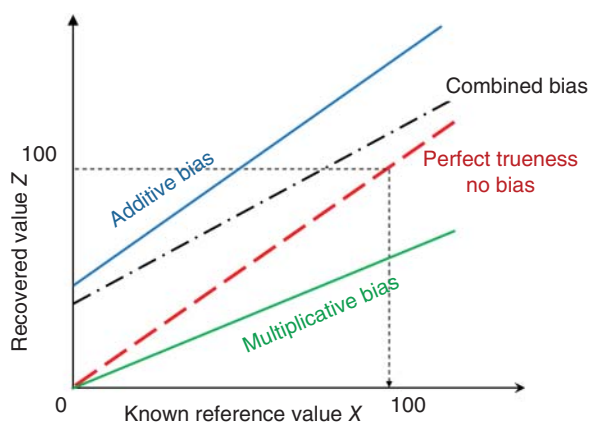


Figure 4.2 Geometric interpretation of additive and multiplicative bias.



analyzed, the inverse-predicted concentrations are reported on the graph for each material. For instance, such a diagram can be obtained with a single material receiving different spikes as explained in Chapter 1 about the standard addition method (SAM). Figure 4.2 illustrates three hypothetical situations:

- Measurement values are systematically upwards shifted in concentration, and the recovery line is shifted by an additive bias (it can be negative, but it still is an additive bias).
- Measurement values are proportionally diverging from the concentration, and the slope of recovery line is modified by a multiplicative bias.
- Both additive and multiplicative biases are combined.

Keeping the same notation, i.e. X the reference value and Z the inverse predicted concentration, these three hypothetical situations can be modeled as follows:

Trueness line (no interferences)	$Z = X$
Multiplicative bias	$Z = b_1 X$
Additive bias	$Z = b_0 + X$
Combined bias	$Z = b_0 + b_1 X$

Unfortunately, it is impossible to experimentally isolate each type of bias, as they are usually combined. But if we assume that the model connecting X and Z is a straight line, the recovery line can be used.

- To check if the trueness is linear or varies in a more complex way with the concentration.
- To estimate the coefficients of the model $Z = b_0 + b_1 X$ and define an eventual correction factor. Examples of correction factors are presented in Sections 8.4.2 and 10.2, with consequences for measurement uncertainty (MU).

When it is not possible to globally evaluate the matrix interferences, the SAM represents a convenient technique to highlight and monitor the interferences; possible procedures are presented in Chapter 1.

Because trueness is defined by reference to a true value, it will be assumed in the rest of this chapter that there is one true value.

4.2 Assessment of Trueness

4.2.1 Primary Operating Procedures

To bring some enlightenment, it is interesting to consider how official metrology bodies and BIPM establish reference values. The BIPM covers nine metrological areas, among them chemistry and biology, i.e. the analytical sciences in general. The other areas are acoustics, ultrasound and vibration, electricity and magnetism, length, mass and related quantities, photometry and radiometry, ionizing radiation, time and frequency, and thermometry. There are related specialized consultative committees that elaborate the way in which the units and the related standards are defined. For the analytical sciences, this is the Consultative Committee for Amount of Substance, or Consultative Committee for the Quantity of Matter (CCQM), one of the latest committees to be created since it was only established in 1993 while other committees already existed.

The difficulty of defining international standards in this area can explain its postponed creation. The task was easier for other committees, while concrete standards, such as the meter or the kilogram, have existed for a long time. To overcome these shortcomings, during the first CCQM meeting in 1993, when chemistry was recognized as a separate field of metrology, it was decided to introduce the primary methods of measurement, defined as a “reference measurement procedure used to obtain a measurement result without relation to a measurement standard for a quantity of the same kind.”

In other words, a primary method of measurement allows a quantity to be measured in terms of a particular International System SI unit without reference to a standard or measure already expressed in that unit. In principle, it is completely independent of the measurements of the same quantity but calls upon measurements expressed in other units of the SI.

Unfortunately, only a few methods of analysis meet that definition. It is mainly coulometry, gravimetry, titrimetric, freezing point depression, and isotope dilution with mass spectrometry (ID-MS). Considering that chromatographic equipment represents more than 80% of the world's instrumental fleet, there is still a long way to go before routine methods are linked to primary methods. As examples of primary standards, VIM cites:

- A primary standard prepared by dissolving a known quantity of a chemical substance in a known volume of solution; typically, a calibration solution.
- A standard for isotope molar ratio measurements, prepared by mixing known amounts of specified isotopes, as a reference material (RM) used for isotopic dilution.

Recently, numerous studies have been launched to validate other methods as primary.

4.2.2 Reference Materials

The difficulty of tracing chemical measurements to SI units, linked to the absence of a unique international material standard for the mole, has been highlighted by CCQM. To overcome this problem, it was decided to develop different RMs. The number of RMs is theoretically infinite, because they are supposed to be available for any molecule in any matrix at various concentrations!

In recent decades, however, there has been a huge increase in their number and interest. Therefore, the increasing number of RMs in analytical chemistry would make an exhaustive list of them obsolete at the time of writing. Nevertheless, several types can be distinguished, depending on their intended use. The strengths and weaknesses of each are summarized in the following list.

4.2.2.1 Certified Reference Materials (CRM)

The reference value is established by a specialized collaborative study involving metrology reference laboratories. The sources of error are carefully controlled and reduced, but despite this, the MU can still be remarkably high compared to the specifications sought. As they are limited in number, their most typical disadvantages are that the matrix used may differ from the one in the scope of the method; they are not available at the desired concentration level; and their price is high and prohibits any daily use.

4.2.2.2 External Reference Materials (ERM)

They have recently appeared on the market. The multiplication of proficiency tests makes it possible to propose the use of surplus samples for trueness verification or calibration purposes. Their reference value and the corresponding uncertainty are obtained at the end of the test, and their traceability can, depending on the case, be traced back to certified reference materials (CRMs). They have the advantage of being more readily available, less expensive than CRMs, and better adapted to the various needs of laboratories.

4.2.2.3 Internal Reference Materials (IRM)

The user determines the reference value. This may be done in collaboration with other laboratories or by combining methods. The matrix used is well suited to the laboratory's field of application, and it is generally cheaper. The sources of error can still be important and their traceability to SI can be insufficient. On the other hand, they are perfectly adapted to control charts.

4.2.2.4 Verification Standard Solutions

They are prepared independently of the calibration solutions. They do not allow the detection of matrix effects. They are independent of the analytical method and inexpensive. They can also be certified either by a metrology organization or by the reagent supplier.

4.2.2.5 Standard Addition Method (SAM) and Surrogate Samples

Both of these methods are described in more detail in Chapter 1 and come with the different available protocols. Surrogate samples can only be considered correct substitutes for internal validation when the nature of the matrix is perfectly known, as in the case of excipients for drugs. They are cheap, but their traceability is often poor.

ISO 17025 requires for an accredited laboratory that, wherever possible, the traceability of RMs to SI units or CRMs be established. Internal reference materials (IRMs) shall be verified to the extent technically and economically feasible. Although CRMs are the most elaborate form of RM, there are several problems associated with their use.

A few years ago, RMs were a rare commodity. Now, many organizations produce or distribute RMs. Around 1995, the Code of Reference Materials or COMAR database was developed by the French National Bureau of Metrology (BNM); today the German BAM, or Bundesanstalt für Materialforschung und -prüfung is managing <https://www.comar.bam.de>. It contains tens of thousands of RM references, manufactured in 27 countries by over 200 producers. The European research program called Virtual Institute for Reference Materials (VIRM) also developed a database of approximately 20,000 CRMs. For a long time, the certification procedure for a CRM was conducted in a non-transparent manner.

The allegation “certified” was therefore used somewhat abusively and had not much to do with the third-party *certification* process defined in the ISO 9000 standards. However, there is no single approach to setting a certified reference value. Often, one or more collaborative studies are performed, but by crossing analytical techniques and selecting a few expert laboratories, to identify major biases. However, some organizations believe that it is preferable to entrust this task to a single highly specialized laboratory that will perform the certification alone.

Of course, the economic aspect is mostly considered when making the choice. Sometimes the recommendations of the ISO 5725 or ISO 13528 standards are far from being followed, even if the statistical exploitation is in agreement with them. Great attention must be paid to the preparation of a CRM, the control of its homogeneity (Section 4.3.5) and its shelf-life. In the case of perishable products, dried or freeze-dried materials are used. For powdered materials, the packaging is made with grain-to-grain filling systems. Sometimes modified-atmosphere packaging is used to stabilize the analytes. An RM must have the following qualities:

- Stability over time.
- Homogeneity of the analyte to be determined.
- Concentration in accordance with the usual applications.
- Availability in sufficient quantity.
- Contain several analytes of interest.

These qualities are particularly critical in the case of biological materials, such as foods or biological fluids. Biological matrices are essentially perishable, and their shelf-life is strictly related to their water content. Unstable analytes such as vitamin D in a plant, since it is photosensitive, has not the same shelf life as traces of cadmium in an alloy. CRMs are intended for use by a single laboratory to conduct an in-house

validation study to verify that their results agree with the certified reference value. Unfortunately, compared to the wide variety of matrices processed by analysts, only a limited number of CRMs are available.

Paradoxically, there is no real recommendation on how to compare the result obtained by the laboratory on a CRM with its certified value. As all these materials are now distributed with the MU on their reference value, a simple method could consist in checking if the result obtained is contained in the coverage interval, including the laboratory's own MU. Section 8.2 on sample conformity assessment gives a detailed explanation of such a comparison procedure.

A preliminary precaution is to check the mode of expression of CRM uncertainty, as standard uncertainty u or an expanded uncertainty U , knowing that, unless otherwise indicated, $U = 2u$. If X is the reference value, $U(X)$ its expanded uncertainty, Z the result obtained by the laboratory, and $U(Z)$ its uncertainty, the following test is a simple decision rule.

Decision rule for CRMs

$$\frac{|Z - X|}{\sqrt{U^2(Z) + U^2(X)}} \leq 1.96 \quad (4.5)$$

When possible, a more efficient method consists of including CRMs as validation materials for building a method accuracy profile (MAP) and directly checking if the bias and tolerance intervals lie in the acceptance interval (Section 5.1). For accreditation bodies, the value of CRMs is unquestionable, so their use is widely recommended. However, in addition to the difficulties inherent in their conservation, many more fundamental analytical problems persist. For example, there are no CRMs for some official methods, such as the Kjeldahl nitrogen method, because its reproducibility is unsuitable for certification. In addition, very often for such direct methods where the analyte is defined by the method, there is no exact molecular equivalence between what is sought and what is measured.

4.3 Proficiency Testing

Fortunately, for those fields of application that do not have CRMs, or do not yet benefit from the work of official metrology organizations nor primary methods, proficiency testing is a possible method for assessing trueness. The classic experimental approach consists of interlaboratory comparison defined as the “organization, performance and evaluation of measurements or tests on the same or similar items by two or more laboratories in accordance with predetermined conditions.” Traceability is achieved to a lesser extent than with CRMS but is quite satisfactory, given the shortcomings. The organization and design of collaborative studies vary according to the objectives. These differences do not always appear clearly in the literature. Two major types of collaborative studies can be distinguished.

4.3.1 Interlaboratory Comparison or Proficiency Testing Scheme (PTS)

It involves many participants (up to several thousand) who do not repeat their measurements. The objective is to rank the laboratories using different scores and verify